3.1 A historical perspective

The idea of representing a signal as a sum of harmonically related sinusoids goes back to the study of vibrating strings and the flow of heat. Fourier claimed that any periodic signal could be written this way; the conditions under which the claim holds were settled later.

3.2 The response of LTI systems to complex exponentials

Complex exponentials are eigenfunctions of linear time-invariant systems: the output for input e^{st} is H(s) e^{st}. A signal written as a combination of complex exponentials therefore passes through such a system term by term.

3.3 Fourier series representation of continuous-time periodic signals

A periodic signal x(t) with fundamental period T can be represented as a linear combination of harmonically related complex exponentials, each with a frequency that is an integer multiple of the fundamental frequency $\omega_0 = 2\pi/T$. The term for k = ±1 is the fundamental, or first harmonic; the terms for k = ±N are the N-th harmonics.

$$x(t) = \Sigma\, a_k\, e^{jk\omega_0 t}, \quad k = -\infty \ldots \infty$$

The coefficients a_k measure how much of each harmonic the signal contains. For the square wave below, only odd harmonics appear.

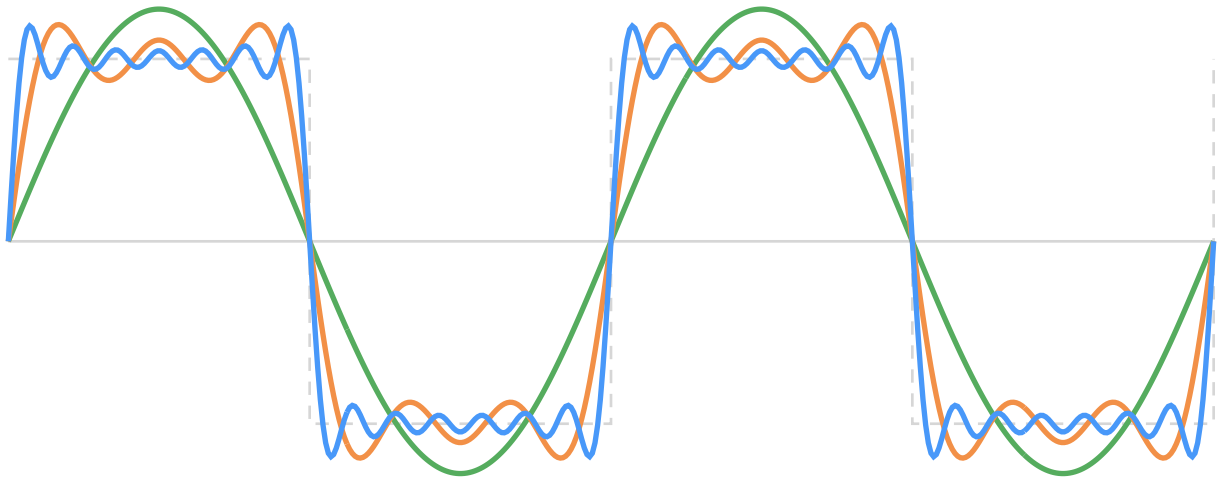

Figure 3.6 Partial sums of the square wave's series with 1, 3 and 7 terms.

3.4 Convergence of the Fourier series

Near a discontinuity, the partial sums overshoot the signal by about 9% of the jump. Adding terms narrows the overshoot but does not lower it: this is the Gibbs phenomenon. The energy in the difference still goes to zero.